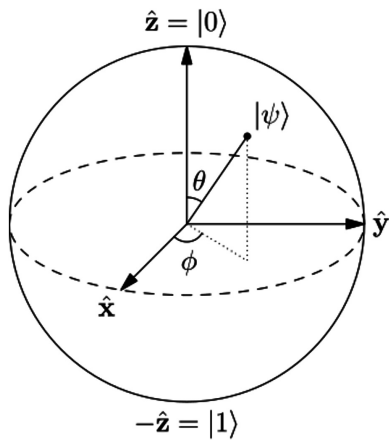


Quantum Coding with Qiskit

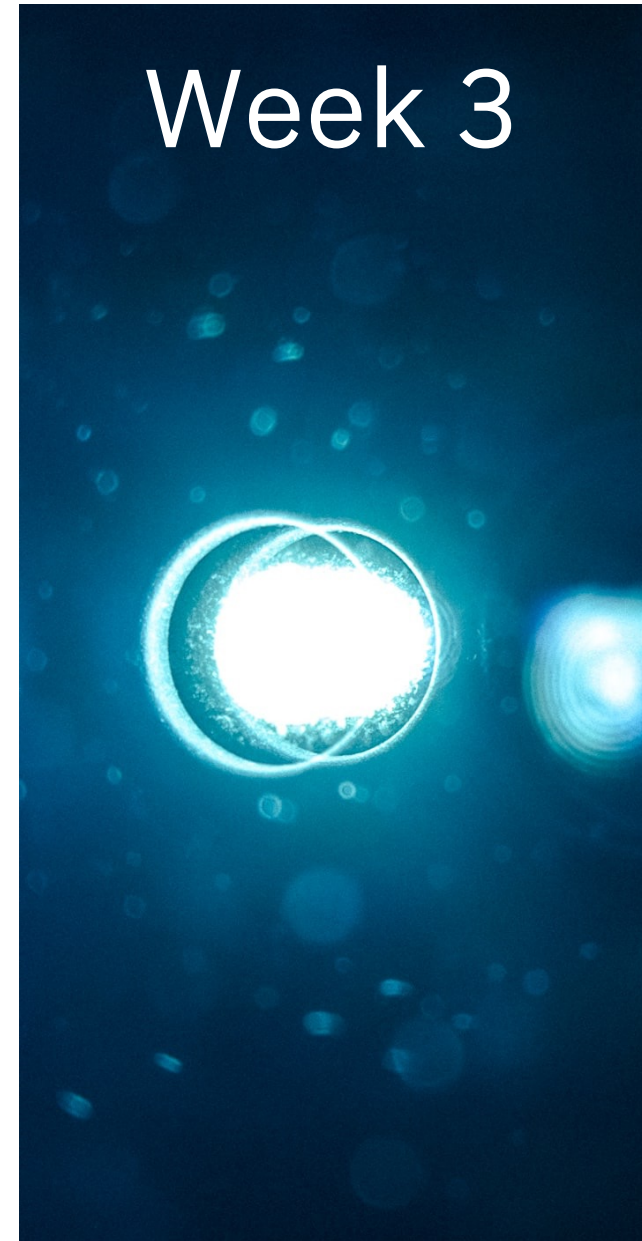
– A Course for Beginners

27th May 2026

Sebastian E. Grodzietzki, Dipl.-Inf.
Joey Rogers, M.Sc. Physics



Week 3



Welcome & Agenda / Week 3

Week 3 — Two qubits, superposition, entanglement 🧐 / This is a **content week**

Recap

Pauli & flexible Rotations

Superposition and Entanglement

- Hadamard gate in depth / Superposition of 2 and n qubits
- CNOT gate in depth / How we entangle

Coding: -> Notebook 2

- Two qubit gates, superposition, entanglement, bell states.
- Preparing superposition and entangled states
- Understanding measuring in different bases
- More visualisation

Q&A

Optional Appendix in the Notebook for deeper dive:

Entanglement vs. Correlations

-> Great chance to interact and/or discuss it with your study group

Before we Start...

Who of you will be onsite in Berlin on 10th June? 🖐️

We'll meet in the
IBM Office at Berlin
Main Train Station

+

Online as always



Week 5 — Flipped Classroom Discussion 🎯

JUN 10 Wednesday, June 10
1:00 PM - 2:30 PM

IBM Deutschland GmbH ↗
Berlin

You have manage access for this event. [Manage](#) ↗

 Presented by
Quantum Coding with Qiskit | A C... >

This is a 3-months Quantum Computing & Qiskit fundamentals course with content weeks & study weeks, focused on hands-on learning Qiskit using the IBM Quantum Platform.

Hosted By

-  **Sebastian Grodzietzki**
-  **Joey Rogers**

Location

IBM Deutschland GmbH
Ella-Trebe-Straße 3, 10557 Berlin, Germany



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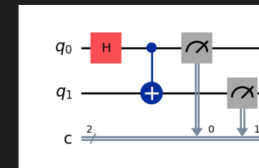
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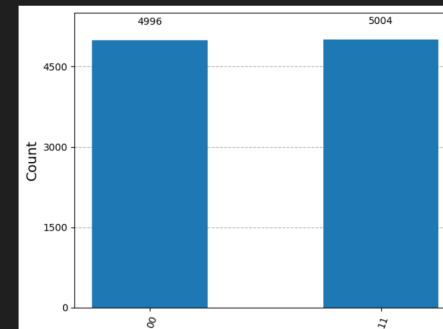
Measuring Entanglement

```
from qiskit_aer import AerSimulator

qc = QuantumCircuit(2, 2)
qc.h(qubit=0)
qc.cx(control_qubit=0, target_qubit=1)
qc.measure(qubit=0, cbit=0, cbits=[0, 1]) # measure qubit(s) [0, 1] into classical bit(s) [0, 1] respectively
qc.draw(output='mpl')
```



```
sim = AerSimulator()
result = sim.run(circuit=qc, shots=10000).result()
counts = result.get_counts()
plot_histogram(data=counts)
```



Q & A

Thank you!!

See you next Wednesday,
same place, 13:00-14:30

Sebastian E. Grodzietzki, Dipl.-Inf.
Joey Rogers, M.Sc. Physics